

Task List 2

1. Implement a custom filter function that replicates the behavior of the native `.filter()` method in JavaScript. The task should involve understanding callback functions and array iterations.
2. Develop a function that can take multiple arrays as input, merge them into a single array, and then sort the merged array in ascending or descending order without using the native `.sort()` method.
3. Create a simplified version of JavaScript's native Promise from scratch, including methods like `.then()`, `.catch()`, and `.finally()`. This will help in understanding asynchronous JavaScript.
4. Write a function that takes a string containing different types of parentheses (`()`, `{}`, `[]`) and checks if they are balanced. Use a stack-based approach to solve this problem.
5. Patterns:

a. $N = 5 \Rightarrow$

```
*****
*       *
*       *
*       *
*       *
*****
```

b. $N = 5 \Rightarrow$

```
      *
    * * *
  * * * *
* * * * *
* * * * *
  * * * *
    * * *
      *
```

c. $N = 10 \Rightarrow$

```

      *
    * *
  *   *
 *     *
*****
      *
    * *
  *   *
 *     *
*****
      *
      *
      *
      *

```

d. $N = 10 \Rightarrow$

```

      A
    B B
  C   C
 D     D
E       E
F       F
E       E
 D     D
  C   C
    B B
      A

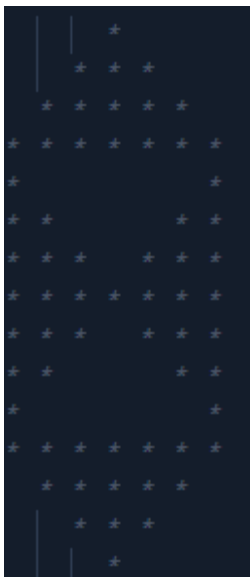
```

e. $N = 10 \Rightarrow$

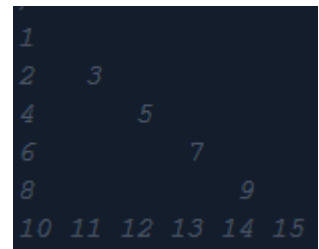
```

ABCDEF
ABCDE
ABCD
ABC
AB
A
A
AB
ABC
ABCD
ABCDE
ABCDEF

```



f. $N = 8 \Rightarrow$



g. $N = 5 \Rightarrow$